

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-YD-2301-07 Sheet 1 of 1
OPL Title	Trunnion Bearing Inspection & Drum Float	Discipline	Mechanical	
Equipment	YD-2301 - POLYMER FLUID BED DRYER	Area / Unit	2300 - POWDER DRYING SYSTEM	
Related Interlock	SEQ-5500 (SIL 2)	P&ID; Ref	TJC-LLD-PID-2301	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **trunnion bearing inspection & drum float** for YD-2301 (POLYMER FLUID BED DRYER) in the POWDER DRYING SYSTEM. The support trunnions carry the full drum weight; worn rollers cause vibration and mis-tracking of the drum. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **Polymer powder / LP Steam** at up to 6 barg (shell) / 150 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Do a listening / vibration check on both trunnions each round	Isolation & LOTO verified
2	Grease the trunnion bearings on the scheduled route	Parts match BOM / spec
3	Check the drum axial float stays within 5 mm	Torque / clearance to spec
4	Inspect the thrust roller for wear and correct tracking	No leak / smooth operation
5	Report any drum wobble or unusual thrust to reliability	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Steam and powder leaking at YD-2301 discharge gland	PTFE 4526L gland packing worn beyond service limit	Replaced 8 pcs PTFE 4526L packing, staggered joints, re-nipped after .
Minor powder weep at feed-end gland	PTFE 4505L packing hardened and worn	Renewed 2 pcs PTFE 4505L feed-end packing
SSLL-2305 speed low alarm, chain slack observed	Chain elongation with normal wear caused slip	Adjusted take-up, lubricated chain, checked sprocket wear

6. KEY LEARNING POINTS

- The support trunnions carry the full drum weight; worn rollers cause vibration and mis-tracking of the drum.
- Respect the interlock SEQ-5500: e.g. TSHH-2301 trips at > 125 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-2301 before working on YD-2301.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
Panel Operator / Technician	Yudha Permana (EMP-1124)	Zulkarnain Hasan (EMP-0901)	Sunday, 26 April 2026

Cross-ref tag YD-2301 across Datasheet, GA, P&ID; Interlock and Maintenance History.